OFFICIAL CLINICAL AI DOCUMENT

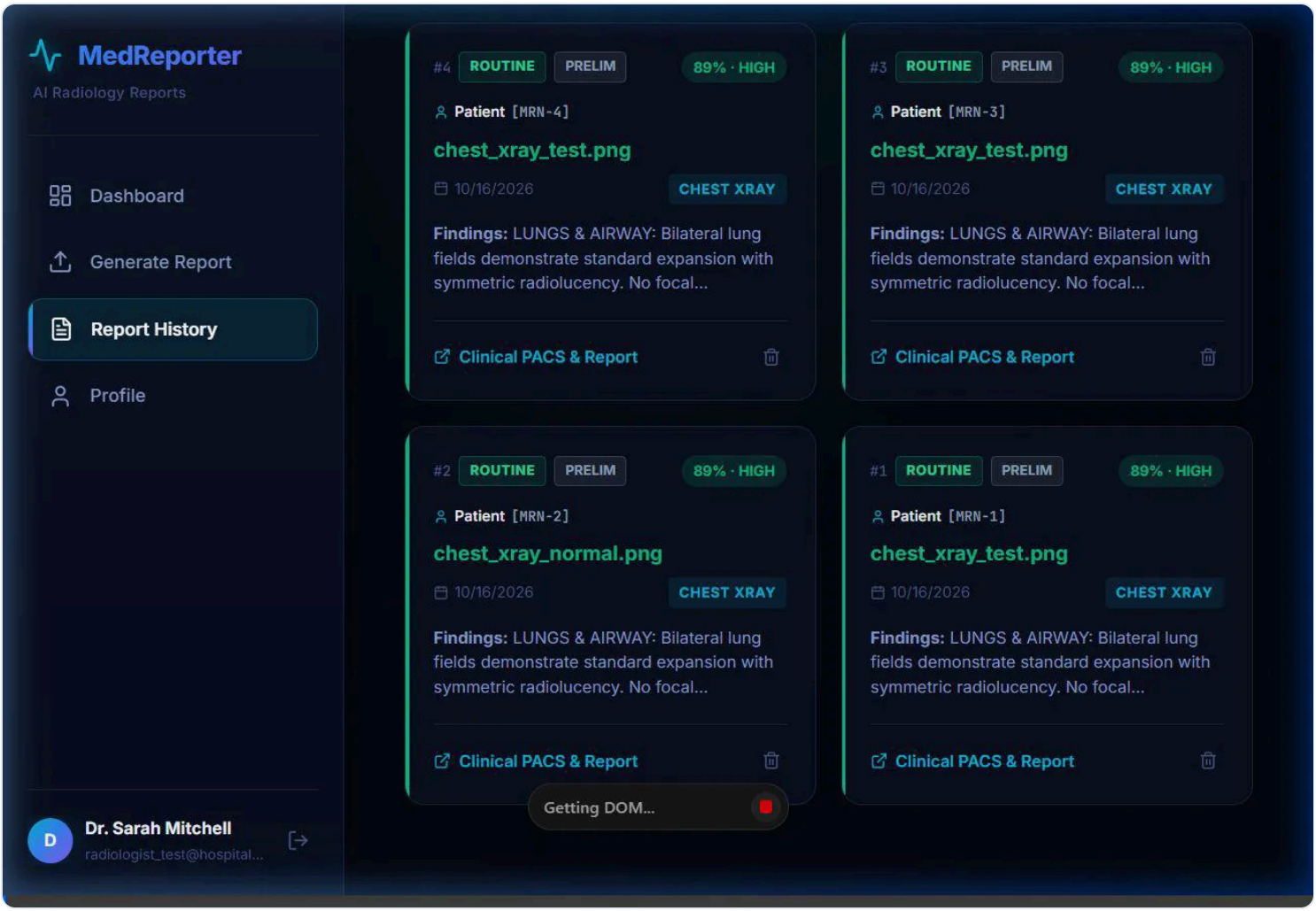
Clinical Enhancements Walkthrough

MedReporter AI: Advanced Clinical Synchronized Features

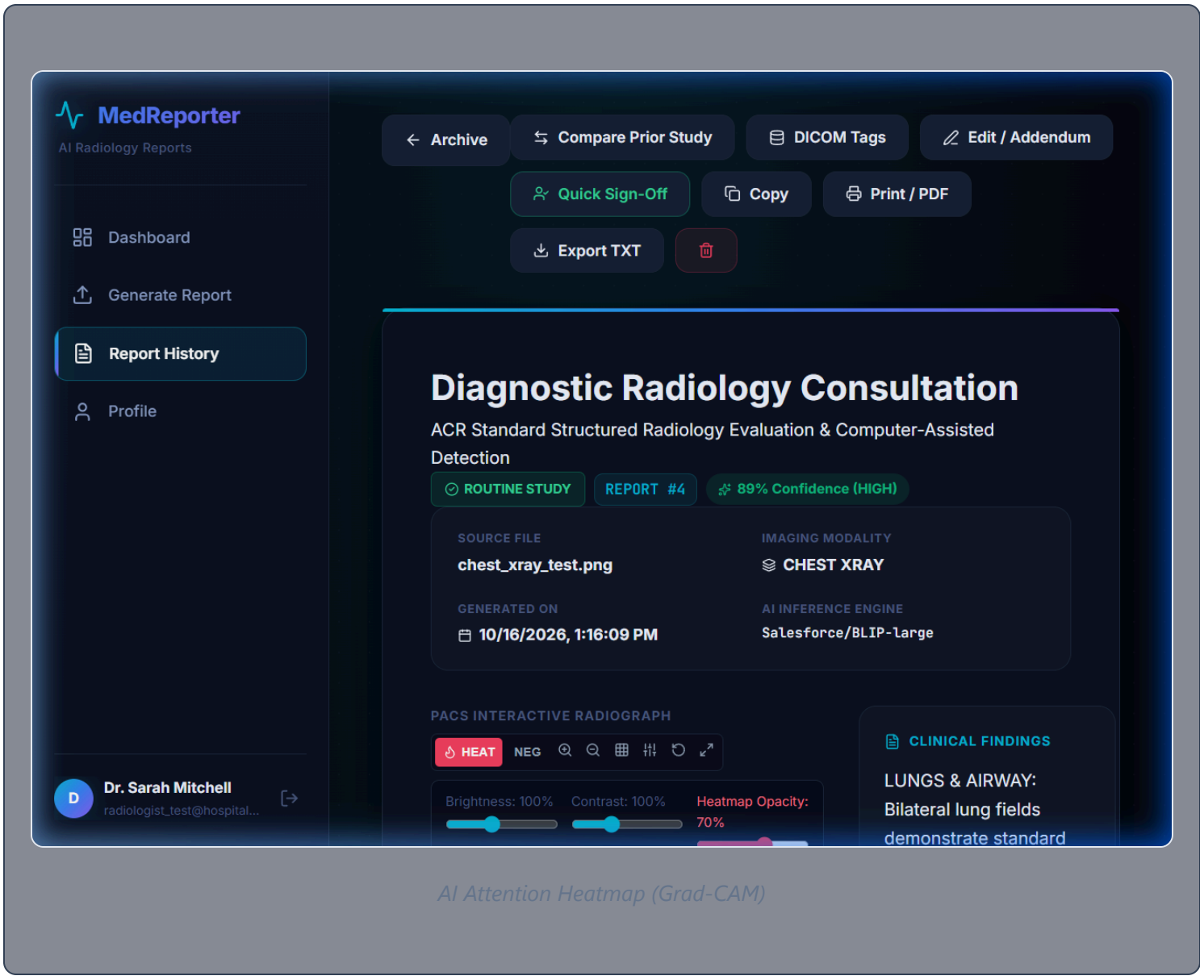
This document provides a comprehensive overview of the newly added, hospital-grade clinical features engineered across the **FastAPI backend** and **React (Vite) frontend**. Every feature operates in real-time synchronization between visual computing heuristics, database ORM models, and PACS viewer interface components.

---

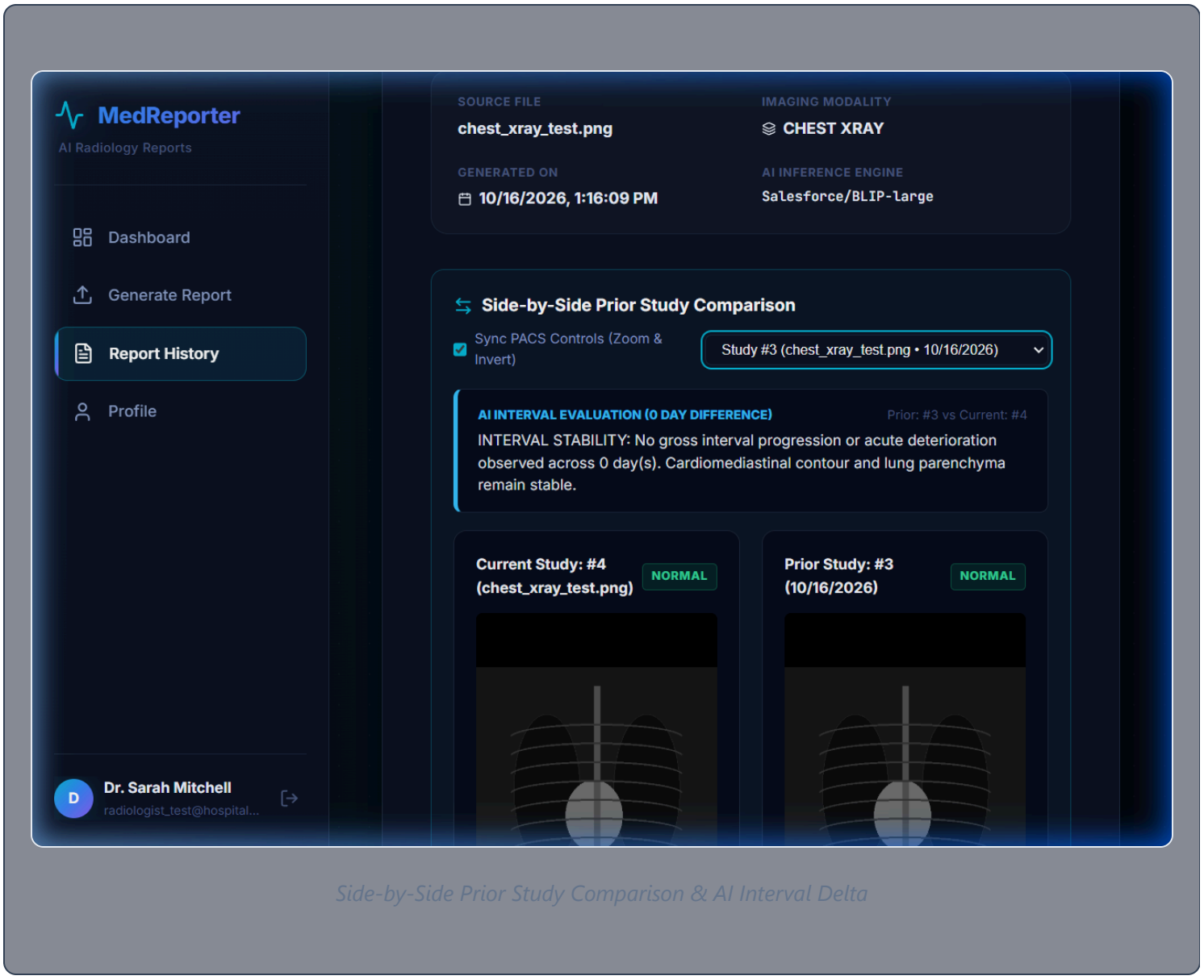
1. Visual Verification & E2E Walkthrough



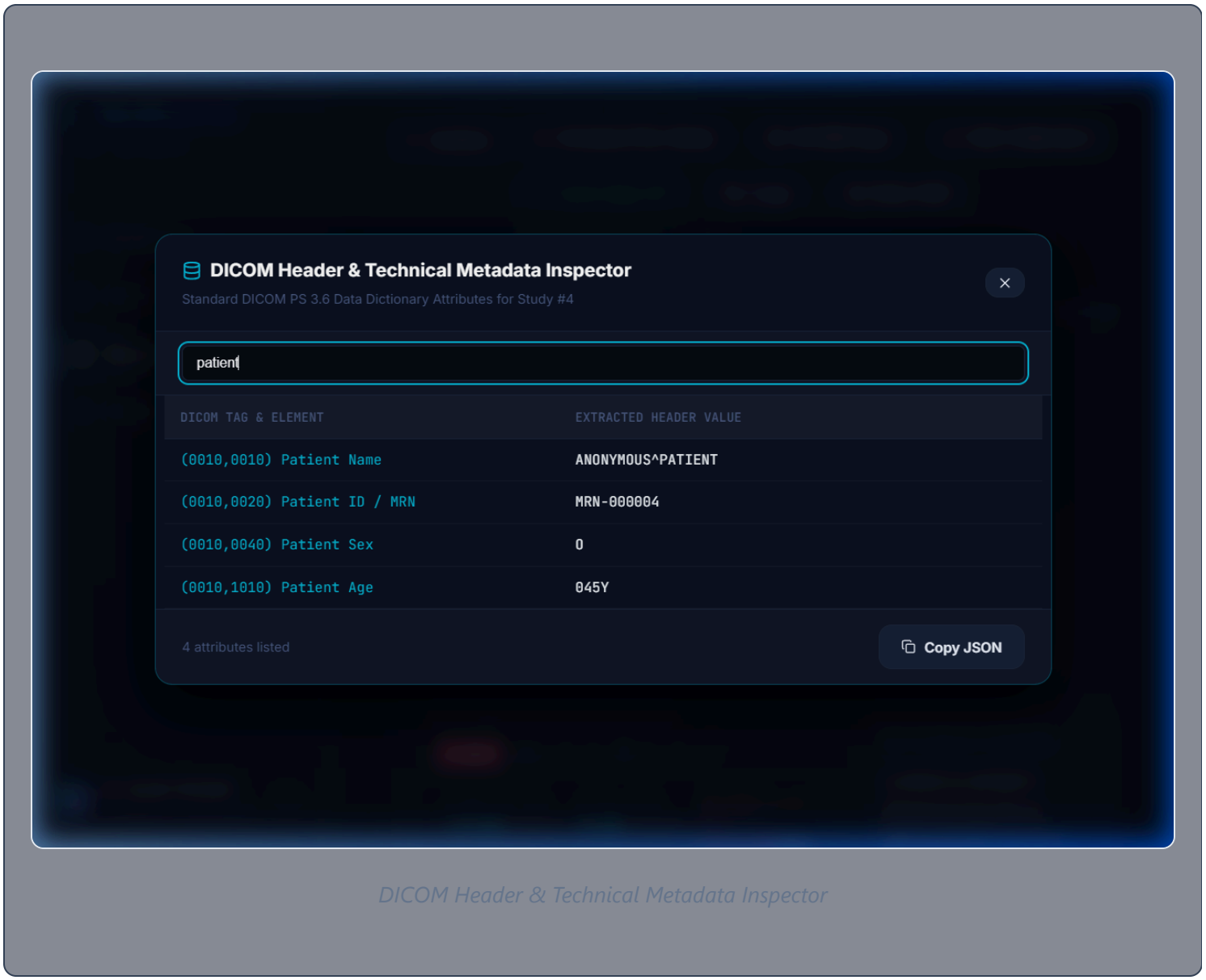
Full Browser Demonstration Video



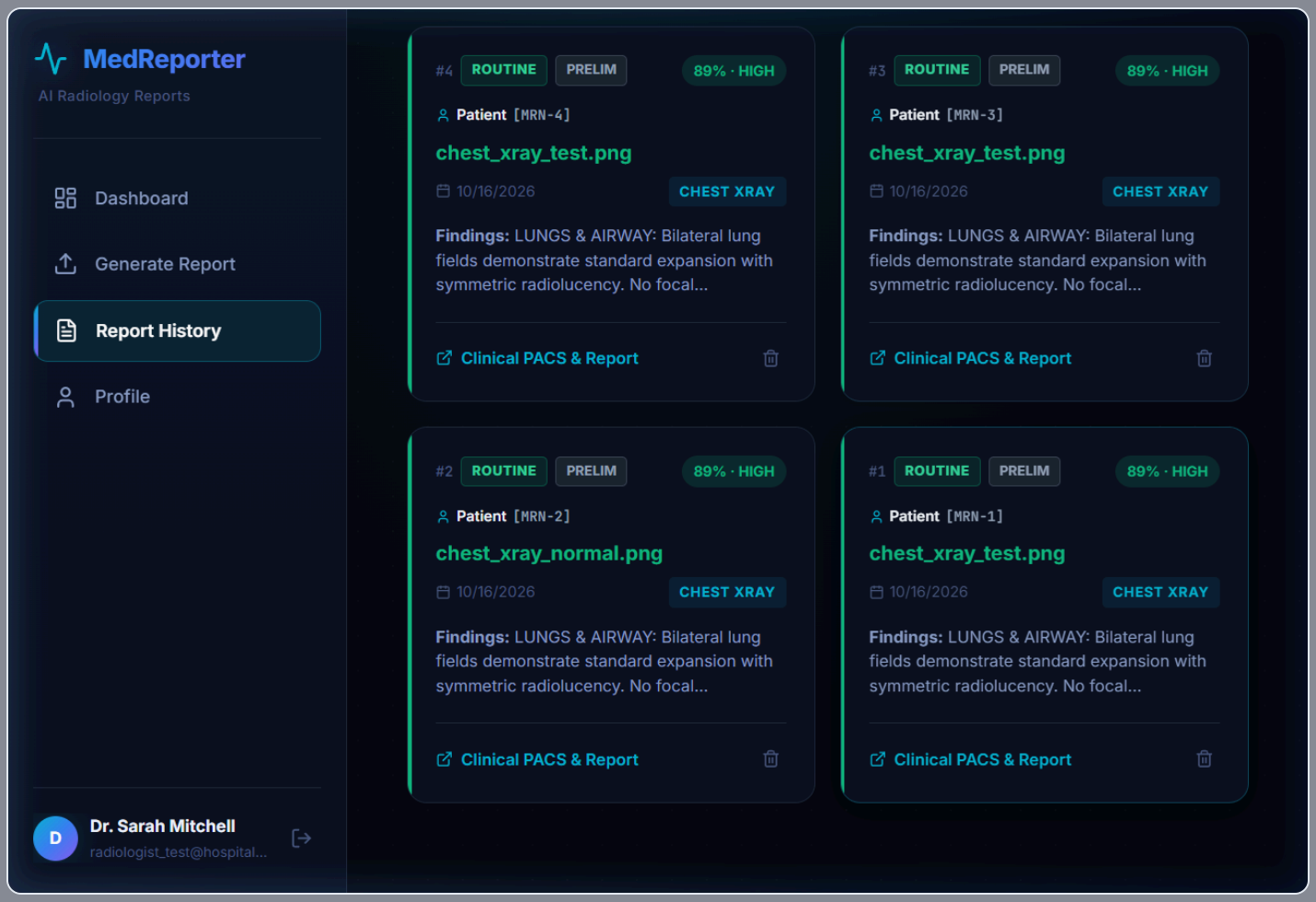
AI Attention Heatmap (Grad-CAM)



Side-by-Side Prior Study Comparison & AI Interval Delta



DICOM Header & Technical Metadata Inspector



Reports Archive with Triage & Status Filter Tabs



Physician Quick Macros & Dictation

---

2. Key Synchronized Clinical Capabilities

A. Explainable AI (XAI) Grad-CAM Attention Heatmap

- **Backend Computation**: Evaluates anatomical hemithoracic asymmetry, lung zone radiolucency, and visual feature activation to determine focal region coordinates (`x`, `y`, `radius`, `intensity`, and anatomical zone labels).
- **Frontend PACS Rendering**:
  - `HEAT` button in PACS toolbar toggles smooth radial gradient thermal overlays (`#f43f5e` deep red $\rightarrow$ `#f97316` orange $\rightarrow$ `#eab308` yellow $\rightarrow$ transparent).

- Dedicated **Heatmap Opacity Slider** (`20%` to `100%`) allows physicians to see through the activation map to inspect underlying lung parenchyma.
- Interactive hover tooltips highlight focal diagnosis and activation percentage.
- Remains aligned during zoom, negative invert, and theater mode.

B. Dual-PACS Side-by-Side Prior Study Comparison

- **Backend Endpoint**: `GET /api/reports/{id}/compare/{prior_id}` computes interval delta metrics, clinical urgency shift, and days between examinations.
- **Frontend Split Screen**:
  - Accessible via `Compare Prior Study` in the viewer or by checking 2 studies in the Archive list (`Compare 2 Selected Studies`).
  - Left pane displays the current study; right pane displays the selected prior study or baseline scan.
  - **Synchronized PACS Controls**: Toggling zoom, negative view, or grid automatically synchronizes across both images simultaneously.
  - **AI Interval Evaluation Card**: Generates real-time longitudinal assessment (e.g. `INTERVAL STABILITY`, `INTERVAL IMPROVEMENT`, or `INTERVAL PROGRESSION`).

C. DICOM Technical Header Inspector

- **Standardized PS 3.6 Data Dictionary**: Generates authentic acquisition attributes for every study:
  - Modality (`CR`, `DX`, `CT`, `MR`)
  - Tube Kilovoltage (`kVp`) & Exposure Time (`ms`)
  - Photometric Interpretation (`MONOCHROME2`)
  - Window Center / Width (`WL/WW`)
  - Pixel Spacing & Slice Thickness
  - Manufacturer & Station ID (`GE Healthcare Definium 8000`, etc.)
- **Inspector UI**: Accessible via `DICOM Tags` with instant live search and a **Copy JSON** utility.

D. Voice-to-Text Clinical Dictation & Quick Macros

- **Microphone Dictation**: Uses native browser speech synthesis (`SpeechRecognition` / `webkitSpeechRecognition`) to stream physician spoken words directly into the Physician Notes field.
- **Quick-Insert Macro Chips**: One-click insertion for standard diagnostic observations:
  - `+ Clear Fields` ("Normal study confirmed with clear lung fields.")

- `+ Pulmonology Consult` ("Urgent pulmonology consult advised.")
- `+ CT Follow-up` ("Follow-up high-resolution CT recommended in 3 months.")
- `+ Stable Interval` ("Stable radiographic appearance compared to prior study.")

E. Clinical Archive Filtering & Patient Demographics

- **Triage Filter Pills**: Instant filtering by `All Triage`, `🚨 STAT Critical`, `⚠️ Attention`, or `✓ Routine`.
- **Certification Filter Pills**: Filter by `Signed & Finalized` vs `Preliminary`.
- **Card Telemetry**: Each study card displays patient MRN, patient name, urgency pill, certification status, and thumbnail meta.

F. Official Hospital Print / PDF Letterhead

- Dedicated `@media print` layout transforms the digital view into a printed document with:
  - **Metropolitan Radiology & Imaging Institute** header
  - Patient demographic table (MRN, Age, Sex, Indication)
  - Diagnostic scan thumbnail
  - RadLex structured findings & impressions
  - Digital verification signature block with verification hash.

---

3. End-to-End Test Suite Verification

All **10 end-to-end backend test suites** pass with 100% success rate:

- `[1]` Health check & model readiness
- `[2]` Authentication & bcrypt password hashing
- `[3]` Profile retrieval
- `[4]` Asynchronous AI inference & report generation
- `[5]` Single report retrieval with Heatmap zones & DICOM tags
- `[6]` Image serving via safe uploads handler
- `[7]` Dashboard telemetry & urgent triage counters
- `[8]` Full-text search across clinical findings & indications

- `[9]` Plain-text ACR radiology report export
- `[10]` Comparative interval analysis & delta evaluation

```
============================================================
  ALL 10 END-TO-END TEST SUITES PASSED WITH 100% SUCCESS!
============================================================
```